

Facility: SALEM 1 & 2Scenario No.: ESG-3Op-Test No.: 11-01 NRC

Examiners: _____

Operators: _____

Initial Conditions: 47%, BOL, Rod control is in manual for insertion of new NI currents, currently performing SC.IC-DC.NIS-0021 for 2N41. 21 Charging pump is C/T due to bio-fouling. 22 charging pump is in service. 23 charging pump is inoperable but available. 21 SGFP O/S for speed control troubleshooting, HDPs O/S, calorimetric just performed SAT IAW requirement of S2.OP-IO.ZZ-0004, Power Operations. 21 and 22 condensate pumps in service. 23A Circulator C/T for corrective maintenance. Minor CW grassing has been occurring.

Turnover: Raise power to 51% @ 10% per hour, place all HDP I/S, continue power ascension to 60% at 10% per hour, all IAW IOP-4.

Event No.	Malf. No.	Event Type*	Event Description
1		N ALL	Raise power
2	AN3855 AN0023 DA003D	CRS	2A1 125VDC batter charger trip (TS)
3	PR0016A	CRS ^I RO	PZR Press Inst fails hi (during first 1% power up) (TS)
4	C325 CN0086B	C CRS PO	Loss of Circ Water Bus Section 23 causes loss of 2 more "A" circulators, degrading Condenser Vacuum
5		R ALL	Power reduction.
6	CN0117A	M ALL	21 Condensate pump trip causes loss of only operating SGFP → Manual Rx trip
7	AF0181B AF0181A B606	C CRS PO	Sequential loss of all AFW→FRHS Red Path
8	CV0208B	C ALL	FRHS Bleed and Feed initiation (no centrifugal charging pumps available) Cond pump recovery.
			CT#1 Establish AFW flow CT#2 Establish RCS Bleed and Feed

* (N)ormal, (R)eactivity, (I)nstrument, (C)omponent, (M)ajor

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: Loss of vacuum, loss of feed, FRHS Cond Pump recovery

SCENARIO NUMBER: 11-01 NRC ESG-3

EFFECTIVE DATE: See Approval Dates below

EXPECTED DURATION: 80 minutes

REVISION NUMBER: 01

PROGRAM:

☐ L.O. REQUAL

☒ INITIAL LICENSE

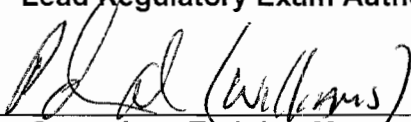
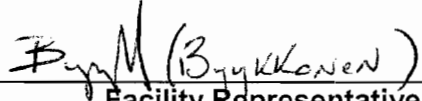
☐ STA

☐ OTHER _____

Revision Summary

New Issue for 09-01 ILOT CERT Exam

Rev 01 6/2012 Replaced CFCU trip with 125VDC Battery charger trip. Added 22 charging pump I/S and subsequent trip to cause immediate Bleed and Feed requirement in FRHS-1.

PREPARED BY:	<u>G Gauding</u> Lead Regulatory Exam Author	<u>8-4-12</u> Date
APPROVED BY:	 Operations Training Manager	<u>9/2-12</u> Date
APPROVED BY:	 Facility Representative	<u>9/11/12</u> Date

I. OBJECTIVES

- A. Given a situation with an OHA alarm, the crew will perform action(s) to investigate and correct the cause of the alarm, IAW approved station procedures.
- B. Given the unit at power with the generator synchronized to the grid, the crew will perform a power increase at 10%/hr, IAW S2.OP-IO.ZZ-0004.
- C. Given the unit at power with a failure of the pressurizer pressure channel, take corrective action IAW S2.OP-AB.PZR-0001.
- D. Given indication of a pressurizer control system malfunction, DIRECT the response to the malfunction in accordance with the approved station procedures.
- E. Given the order or indications of a pressurizer control system malfunction, perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with the approved station procedures
- F. Given the order or indications of a feedwater or condensate system malfunction, perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with the approved station procedures.
- G. Given indication of a feedwater or condensate system malfunction, DIRECT the response to the malfunction in accordance with the approved station procedures
- H. Given a loss of a heater drain pump or condensate pump, take corrective action IAW S2.OP-AB.CN-0001.
- I. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with the approved station procedures.
- J. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with the approved station procedures.
- K. Given the order or indications of a loss of secondary heat sink, perform actions as the nuclear control operator to RESPOND to the loss of heat sink in accordance with the approved station procedures.
- L. Given indication of a loss of secondary heat sink, DIRECT the response to the heat sink loss in accordance with the approved station procedures.
- M. During performance of emergency operating procedures, monitor the critical safety function status trees in accordance the EOP in effect.

II. MAJOR EVENTS

- A. PZR Pressure controlling channel fails high
- B. Loss of 2 additional circulators/vacuum degradation.
- C. Loss of only operating SGFP >P-10.
- D. FRHS with no charging pumps.
- E. FRHS Cond pump recovery.

III. SCENARIO SUMMARY

- A. The crew will take the watch with Rx power at 48%, BOL. Power was reduced yesterday during CW grassing. 21 SGFP developed speed oscillations during the downpower and was removed from service for troubleshooting. 23A Circulator is C/T for Traveling Screen corrective maintenance. Rod control is in manual for insertion of new NI currents, currently performing SC.IC-DC.NIS-0021 for 2N41, which is the last of the 4 channels to be updated. The adjustment is complete, and technicians are finishing the paperwork. All Heater Drain Pumps are O/S. 21 and 22 condensate pumps in service. Orders for the shift are to raise power to 51% at 10% per hour, place HDPs in service, then continue power ascension to 60%. The Rx fuel is conditioned to 90% power.
- B. Shortly after the crew assumes the watch, the 2A1 125VDC Battery Charger will trip. The CRS will enter the appropriate Tech Spec, and dispatch an operator to investigate.
- C. After initiating the power ascension to 51%, the controlling PZR pressure channel will fail high. Operators will enter S2.OP-AB.PZR-0001, Pressurizer Pressure Control Malfunction, stabilize the plant, place a redundant channel in control, and return the Master Pressure Controller to automatic. The CRS will enter the appropriate Tech Spec.
- D. After power has been raised, a loss of 2CW 4KV bus section 23 occurs, and the 2 remaining "A" circulators trip. The CRS will enter S2.OP-AB.CW-0001, Circulating Water System Malfunction. Condenser vacuum will begin to degrade, and the crew will commence a power reduction.
- E. During the power reduction as vacuum continues to degrade, 21 condensate pump will trip. The remaining condensate pump will be unable to supply sufficient flow, and as 22 SGFP speed rises, its suction pressure will lower rapidly to the SGFP trip setpoint. The crew will manually trip the Reactor upon the loss of the only operating SGFP.
- F. The CRS will transition to TRIP-2, Rx Trip Response, after the immediate actions of TRIP-1 are performed and SI is not required. No AFW flow will be present following the Rx trip. 22 AFW pump trips after starting, 21 AFW pump pressure override circuit fails, and 23 AFW pump does not auto start. The crew starts 23 AFW pump, and 21 AFW pump trips after its pressure override circuit is defeated.
- G. 5 minutes after the Rx trip, the operating centrifugal charging pump (22) will trip, and letdown will automatically isolate. After establishing AFW flow, the last remaining AFW pump (23) develops a speed control problem which causes its speed demand to lower speed to zero. With FRPs in effect, the CRS will transition to FRHS-1, Loss of Secondary Heat Sink with AFW flow < 22E4 lbm/hr and Steam Generator Narrow Range levels all <9%.
- H. The crew will establish RCS Bleed and Feed based on having no centrifugal charging pumps. The crew will depressurize a steam generator and establish Condensate flow to the depressurized steam generator.
- I. The scenario will terminate when feed flow has been established to at least one SG, and SG WR level(s) are rising.

PSEG Restricted – Possession Requires Specific Approval from Nuclear Training

IV. INITIAL CONDITIONS

___ Pre-snapped IC-243

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

<i>Initial</i>	Description
___ 1	RH1 and RH2 C/T
___ 2	RH18s C/T
___ 3	VC1and VC4 C/T
___ 4	RCPs (SELF CHECK)
___ 5	RTBs (SELF CHECK)
___ 6	MS167s (SELF CHECK)
___ 7	500 KV SWYD (SELF CHECK)
___ 8	SGFP Trip (SELF CHECK)
___ 9	23 CV PP (SELF CHECK)
___ 10	23A Circulator C/T
___ 11	21 SGFP O/S
___ 12	Place copy of SC.IC-DC.NIS-0021, Exhibit 1 on NCO desk.
___ 13	22 charging pump in service
___ 14	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
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	1	EVENT ACTION: COMMAND: PURPOSE:	MONP254 <10 // CONT ROD BANK C < 10 (RX TRIP) DMF CN0086B <update as needed>
	5	EVENT ACTION: COMMAND: PURPOSE:	kb507azp // 21 AUX FEED PUMP-PRESSURE OVERRI <update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
01	PR0016A PZR PRESS CH I (PT455) FAILS H/L	N/A	N/A	N/A	RT-3	2500
02	CN0117A 21 CONDENSATE PUMP TRIP	N/A	N/A	N/A	RT-7	
03	AF0181B 22 AUX FEEDWATER PUMP TRIP	00:00:29	N/A	N/A	ET-1	
04	AN0304 SER 304 FAILS - :A4 REACTOR PROTECTION SYS CH I INSTRUMENT LOOP IN TEST	N/A	N/A	N/A	N/A	SER POINT FAILS/OVRD TO ON
05	CW0114F 23B CIRCULATING WATER PUMP TRIP	N/A	0.3	00:03:00	RT-5	0.9
06	CW0114D 22B CIRCULATING WATER PUMP TRIP	N/A	0.3	00:05:00	RT-5	0.6
07	CW0114C 22A CIRCULATING WATER PUMP TRIP	N/A	N/A	N/A	RT-5	0
08	CW0114A 21A CIRCULATING WATER PUMP TRIP	N/A	N/A	N/A	RT-5	0
09	AF0182A 21 AFP PRESS OVRD PROT FAILS	N/A	N/A	N/A	N/A	
10	RP0344 23 AFW PUMP FAILS TO AUTO START	N/A	N/A	N/A	N/A	
11	VL0031 23CN79 Fails to Position (0-100%)	N/A	N/A	N/A	N/A	100
12	CW0114E 23A CIRCULATING WATER PUMP TRIP	N/A	N/A	N/A	N/A	0
13	AN3855 AAS 855 FAILS - :2A1/2A2 125VDC BATTERY CHARGERS "TROUBLE"	N/A	N/A	N/A	RT-1	AAS POINT FAILS/OVRD TO ON
14	AN0023 SER 023 FAILS - :B2 2A 125VDC CONTROL BUS VOLTAGE LOW	00:00:03	N/A	N/A	RT-1	SER POINT FAILS/OVRD TO ON
15	CN0086B Loss of 22 main condenser vacuum (thor load)	00:01:00	17	00:07:00	RT-5	20
16	AF0181A 21 AUX FEEDWATER PUMP TRIP	00:00:05	N/A	N/A	ET-5	
17	CV0208B 22 CHARGING PUMP TRIP	00:05:00	N/A	N/A	ET-1	

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
01	RC01D OT DELTA T TRIP CH I BS (411C)	N/A	N/A	N/A	N/A	TRIP
02	RC05D OT DELTA T R/BCK CH I BS (411D)	N/A	N/A	N/A	N/A	TRIP
03	CW03A 23A CIRC WATER PUMP CONTROL PWR BKR 2CW4AD	N/A	N/A	N/A	N/A	OFF
04	DA003D BATTERY CHARGER A1 STATUS	N/A	N/A	N/A	RT-1	OFFLINE
05	RC05A RCS SYSTEM , BORON CONC RESET	N/A	N/A	N/A	N/A	1365.1
06	CV44D 21 CHG PUMP BKR CONTROL POWER	N/A	N/A	N/A	N/A	OFF

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
___ 01	B526 L AO 21 STM GEN FEED PUMP SPEED	N/A	N/A	N/A	N/A	55.5
___ 02	B524 A1 LO 21 SGFP TURNING GEAR ENGAGE	N/A	N/A	N/A	N/A	ON
___ 03	B524 A2 LO 21 SGFP TURNING GEAR DISENGAGE	N/A	N/A	N/A	N/A	OFF
___ 04	C325 F DI 23CW1AD OPEN	N/A	N/A	N/A	RT-5	ON
___ 05	B606 F DI 23 AUX FEED PUMP-SPEED DECREASE	N/A	N/A	N/A	RT-9	ON
___ 06	5811 A LO 23A CIRC FORWARD LOW SPEED	N/A	N/A	N/A	N/A	OFF
___ 07	5811 D LO 23A CIRC LOSS OF POWER	N/A	N/A	N/A	N/A	ON
___ 08	CH04 E DI 23 CONDENSATE PUMP-START	N/A	N/A	N/A	N/A	OFF
___ 09	C328 E DI 2CW2BD CLOSE	N/A	N/A	N/A	N/A	OFF

OTHER CONDITIONS:

	Description
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___ 1.

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Power Ascension Note: Digital EHC is set up initially for 1% per minute. PO must set it correctly for 10% per hour. Proceed to next event after a noticeable change in turbine and Rx power has occurred.			
	CRS direct RO to maintain Tave/Tref in a designated band using control rods in manual.		
	CRS directs RO and PO to coordinate actions while raising power.		
	PO initiates turbine load increase to 51% power at 10% per hour.		
	RO withdraws control rods to maintain Tavg/Tref within designated band.		
2. 2A1 125 VDC Battery Charger Trip			
Simulator Operator: Insert RT-1 on direction from Lead Evaluator. After power ascension is started. MALFS: AN3855 2A1/2A2 Battery Charger Trouble AN0023 SER 023 FAILS - :B2 2A			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
125VDC CONTROL BUS VOLTAGE LOW Remote: DA003D BATTERY CHARGER A1 STATUS OFFLINE Role Play: 2 minutes after being dispatched, report as NEO that the supply breaker on the 2A1 125 VDC Battery Charger is tripped. Report there is no apparent reason for the breaker trip.			
	RO reports OHA A-41 AUX ALM SYS PRINTER and OHA B-2 2A 125VDC CNTRL BUS VOLT LO as unexpected.		
	PO reports point in alarm is for 2A1/2A2 125 VDC Battery Charger Trouble.		
	PO refers to ARP for OHA B-2 and reports low bus voltage alarm setpoint is <128 VDC, and current reading is 127 VDC.		
	CRS dispatches an operator to investigate.		
	CRS contacts WCC / Maintenance to investigate.		
	CRS enters TSAS 3.8.2.3 action b.		
	CRS determines that the 2A2 125 VDC Battery Charger must be placed in service within the next 2 hours if 2A1 Battery Charger cannot be restored to operable status.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Continue to next event after Tech Spec determination is made or at direction of Lead Evaluator.			
3. Controlling PZR Pressure Channel I (PT-455) fails high			
Simulator Operator: Insert RT-3 on direction from Lead Evaluator.			
MALF: PR0016A PZR PRESS CH I (PT455) FAILS H/L			
	RO announces RC PRESS DEVIATION HI console alarm, OHA D-8 RC PRESS HI and OHA E-42 2PR1/2 TRIP as unexpected.		
	RO diagnoses PZR Pressure channel I failed high, and requests permission to place MPC in manual.		
	CRS directs RO to place MPC in manual and raise PZR pressure.		
	RO places MPC in manual and lowers demand to minimum, and announces that control has been established over MPC output.		
	RO reports RC PRESS DEVIATION HI and OHA E-28 cleared as expected when reducing MPC demand, and 22 BU PZR heaters		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	energized as expected.		
	CRS enters S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction.		
	CRS directs the power ascension put on hold.		
	PO places the Main Turbine on hold.		
	Crew maintains positive control over reactivity.		
	CRS directs initiation of Attachment 1 CAS of S2.OP-AB.PZR-0001.		
	PO reports POPS is not in service.		
	PO reports controlling PZR Pressure channel I is failed high.		
	PO reports PZR Master Pressure Controller is in manual with demand signal consistent with Attachment 2 and is restoring PZR pressure to program value.		
	PO selects PZR pressure channel III for control.		
	PO returns Master Pressure Controller to auto.		
	CRS direct removal of failed channel from service IAW S2.OP-SO.RPS-0003, Placing Pressurizer Channel in Tripped Condition.		
	CRS enters TSAS 3.3.1.1 Act 6, 3.3.2.1 Act. b		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Do <u>NOT</u> remove power from 2PR6 during scenario. <u>If asked</u> later in scenario why the power has not been removed, reply that the breaker would not open and electrical has been dispatched to investigate. Proceed to next event after power ascension has re-started or on direction from Lead Evaluator. If CRS takes no action to perform the power ascension, then call as Rob DeNight (SOS) and direct the CRS to perform the power ascension.	19, 3.2.5 (DNB when applicable) and 3.4.5.b		
	RO shuts 2PR6.		
	CRS dispatches an operator to remove power from 2PR6.		
	CRS directs RO and PO to proceed with power ascension.		
	PO initiates turbine load increase to 51% power at 10% per hour.		
	RO withdraws control rods to maintain Tavg/Tref within designated band.		
	RO/PO announces when Below P-9 light extinguishes.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
4. Loss of 2A 4KV Circ Water Bus			
Simulator Operator: Insert RT-5 on direction from Lead Evaluator. OVDI: C325 23CW1AD Open Final Value: ON MALFS: CN00086B Loss of Main Condenser Vacuum Ramp: 7 min Final Value: 20 Delay: 1 min CW0114F 23B Circulator Trip Ramp: 3 min Final Value: 0.9 CW0114D 22B Circulator Trip Ramp: 5 min Final Value: 0.6 CW0114C 22A Circulator Trip Final Value: 0.0 CW0114A 21A Circulator Trip Final Value: 0.0			
	PO announces OHAs K-27 23 STA XFMR TRBL and K-34 4KV CW BUS UNDRVOLT as unexpected.		
	PO diagnoses loss of 2CW 4KV bus section 23.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Main Condenser back pressure is limited to 5.0" with Main Generator load <560 MWe.	CRS enters S2.OP-AB.CW-0001, Circulating Water System Malfunction based on 2 or more circulators out of service.		
	PO refers to ARPs for the K overhead windows.		
	CRS directs initiation of Attachment 1 CAS of S2.OP-AB.CW-0001.		
	PO reports there is no indication of a pipe rupture.		
	PO reports 3 circulators are out of service.		
	PO reports at least one circulator is operating on each condenser.		
	CRS dispatches operators to open hood spray bypass valves and monitor condensate pump suction piping.		
	PO reports condenser backpressure is rising slowly.		
	Simulator Operator: Adjust MALF CN0086B as necessary to ensure condenser vacuum rises enough to make crew initiate a power		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Simulator Operator: Insert RT-7 on direction from Lead Evaluator MALF: CN0117A 21 CONDENSATE PUMP TRIP			
Note: If crew attempts to start the idle condensate pump it will not start.	PO announces trip of 21 condensate pump.		
	CRS enters S2.OP-AB.CN-0001, Main Feedwater/Condensate System Abnormality.		
	PO reports lowering SGFP suction pressure.		
	PO reports SGFP Condensate Suction Pressure Lo Console alarms.		
	PO reports 22 SGFP speed is increasing and SGFP suction pressure continues to lower.		
Note: SGFP Condensate Suction Pressure Lo Console alarms at 275 psig.			
Note: SGFP trip on low suction pressure will occur at: <215 psig sustained for 10 seconds <u>or</u> 190 psig with no time delay and will auto start both MDAFW	PO reports when 22 SGFP trips.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
pumps.			
	CRS directs the RO to trip the Rx.		
Manual Rx trip is based on action of AB.CN with no SGFPs running and Rx power >10%.			
	RO initiates a manual Rx trip.		
Simulator Instructor: Ensure ET-1 is TRUE when the Rx is tripped. This deletes the lowering cond vacuum, trips 22 AFW pump and trips 22 charging pump after 5 minute delay.			
	RO performs Immediate Actions for TRIP-1, Reactor Trip or Safety Injection: - Trip reactor - Confirms reactor trip - Trips turbine - Verifies 4KV vital buses energized - Reports SI not demanded or required		
	CRS and RO verify performance of immediate actions.		
	CRS transitions to TRIP-2, Reactor Trip Response, based on no SI actuated or required.		
	Crew begins monitoring CFSTs.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports no AFW flow, and the status of AFW: - 21 AFW pump running with no flow and adequate discharge pressure - 22 AFW pump is tripped - 23 AFW pump has not started		
	CRS directs PO to start 23 AFW pump, and defeat Pressure Override for 21 AFW pump.		
	PO starts 23 AFW pump and reports it is starting. Depresses 21 AFW pump pressure override defeat and reports 21 AFW pump tripped shortly after AFW flow initiated.		
Simulator Operator: Ensure ET-5 is TRUE when Pressure Override Defeat is depressed for 21 AFW pump. This trips 21 AFW pump after 5 second delay.			
CT#1 (TRIP-2 Basis) Establish the minimum required AFW flow (22E4 lbm/hr) prior to a transition to FRHS-1 being required. SAT UNSAT			
	RO makes page announcement.		
	PO adjusts 23 AFW pump speed or 21-24 AF11s to lower total AFW flow to no less than 22E4 lbm/hr.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports AFW flow is > 22E4 lbm/hr.		
	PO reports both SGFPs are tripped.		
Simulator Operator: Insert RT-9 after PO reports both SGFPs are not running. OVERRIDE: B606 F DI 23 Aux Feed Pump-Speed Decrease Condition: ON			
7. Loss of all AFW flow Note: The STA will not arrive in the control room until 10 minutes after being paged. The crew is responsible for identifying the Heat Sink RED path. Depending on scenario flow, SGNR levels may not be <9% when the last AFW pump is lost. The crew should continue actions in TRIP-2 until entry conditions for FRHS-1			
	PO reports 23 AFW pump speed lowering and no AFW flow.		
	RO reports 22 charging pump trip.		
	The crew identifies when a Heat Sink RED path exists when all SG NR levels are <9% and no feed flow exists.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
are met. (AFW flow <22E4 lbm/hr and all SG NR levels <9%.)			
8. RCS Bleed and Feed	CRS transitions to FRHS-1, Loss of Secondary Heat Sink.		
	PO reports it was not operator action which caused total AFW flow to lower < 22E4 lbm/hr.		
	RO reports RCS pressure is > all SGs, and RCS Thots are > 350°F.		
	RO reports no centrifugal charging pumps are available.		
	CRS goes to Step 23 Bleed and Feed initiation.		
	RO stops all RCPs.		
	RO initiates Safety Injection.		
	PO reports SI valves in safeguards position per Table B.		
	RO reports 21 and 22 charging pumps are not running.		
	RO reports both SI pumps are running.		
	RO reports valves in Table C are in correct position.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
CT#2B (FR-H.1 --B) Establish RCS Bleed and Feed before PZR PORVs open automatically due to loss of heat sink. SAT UNSAT	RO reports 2PR6 is shut with power available, and 2PR7 is open.		
	RO opens 2PR6.		
	RO opens both PZR PORVs.		
	RO reports both PZR PORV Stop Valves and both PZR PORVs are open.		
	PO verifies SI valve alignment per APPX-3.		
	RO reports safeguards have not been reset.		
	RO resets SI and Phase A isolation, and reports Phase B is reset.		
	RO opens 21 and 22CA330s		
	PO resets all SECs and reports all SECs and 230V control centers are reset.		
	Crew maintains bleed and feed.		
	RO reports containment pressure remains <15		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Steam dump availability is based on having condenser vacuum, at least one circulator in service on a waterbox and no MSLI initiated. Note: IF CRS determines steam dumps are not available due to previous vacuum problem, then the PO will initiate MSLI on ALL loops and fully open the selected SG MS10 to depressurize selected SG.	psig.		
	RO reports no containment spray pumps running.		
	CRS continues attempts to restore heat sink.		
	CRS determines no AFW is available, and Main Feed is unavailable.		
	PO reports condensate system is in operation.		
	Crew selects only one SG for depressurization.		
	CRS reviews depressurization termination criteria.		
	PO reports Steam Dumps are available.		
	RO initiates MSLI for all loops EXCEPT for		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Simulator Operator: 2 minutes after being directed, use following remote to open the selected 22BF40 or 24BF40. REMOTE: BF06A, 22BF40 fails to position (0-100%) Final Value: 100 Ramp: 00:00:45 REMOTE: BF08A, 24BF40 fails to position (0-100%) Final Value: 100 Ramp: 00:00:45	selected SG.		
	PO places steam dumps in manual, adjust steam pressure valve demand to zero, then places steam dumps in MS Pressure Control.		
	PO adjusts valve demand to 25%, and bypasses Tavg when Tavg lowers to 543°F.		
	CRS dispatches operator to open selected SG BF40.		
	PO opens selected BF13.		
	PO releases or verifies released selected BF22.		
	PO opens 21 and 22 CN48s and closes 21 and 22 CN32s.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Depressurization of selected SG should not be stopped until WR level is rising in selected SG.			
	PO reports indication of feed flow to selected SG and WR level rising.		
The Scenario can be terminated at the Lead Evaluator's discretion, after SG WR level is observed to be rising.			

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction
- F. S2.OP-AB.CW-0001, Circulating Water System Abnormality
- G. S2.OP-AB.COND-0001, Loss of Condenser Vacuum
- H. S2.OP-AB.LOAD-0001, Rapid Load Reduction
- I. S2.OP-AB.CN-0001, Main Feedwater/Condensate System Abnormality
- J. S2-EOP-TRIP-1, Rx Trip or Safety Injection
- K. 2-EOP-TRIP-2, Rx Trip Response
- L. 2-EOP-FRHS-1, Loss of Secondary Heat Sink

ATTACHMENT 1 UNIT TWO PLANT STATUS TODAY

MODE: 1 POWER: 47% RCS BORON: 1365 MWe 500

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Xenon building in at 2 pcm per hour.

Reactivity plan for power ascension is control rods only for ascension to 60%.

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

21 charging pump C/T for biofouling. TSAS 3.5.2.a and 3.1.2.4 (23 charging pump also inoperable) both 70 hours remaining.

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

Raise power to 51% at 10% per hour, place HDPs in service, then continue power ascension to 60% IAW S2.OP-IO.ZZ-0004, Power Operation.

Currently performing SC.IC-DC.NIS-0021 for 2N41, which is last NI channel to be updated. New data is installed, two bistables (411C and 411D) have not been restored yet, and procedure should be finished within next 15 minutes.

ABNORMAL PLANT CONFIGURATIONS:

Power was reduced yesterday during CW grassing. Rod control is in manual for insertion of new NI currents. 22 charging pump is in service to allow troubleshooting of 23 charging pump speed changer problem.

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

No penalty minutes in the last 24 hrs.

PRIMARY:

23 charging pump is O/S and available.

Power Range channel 2N41, 3.3.1.1 Action 2.

SECONDARY:

21 SGFP developed speed oscillations during the down power and was removed from service for troubleshooting. Currently on turning gear.

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

23A Circulator is C/T for emergent Traveling Screen repair

PSEG Restricted – Possession Requires Specific Approval from Nuclear Training

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
with Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.

ATTACHMENT 3

CRITICAL TASK METHODOLOGY

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4
SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 11-01 NRC ESG-3 **REVIEWER:** E Gallagher

Initials	Qualitative Attributes
EG	1. The scenario has clearly stated objectives in the scenario.
EG	2. The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
EG	3. The scenario consists mostly of related events.
EG	4. Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
EG	5. No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
EG	6. The events are valid with regard to physics and thermodynamics.
EG	7. Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
EG	8. The simulator modeling is not altered.
EG	9. All crew competencies can be evaluated.
EG	10. The scenario has been validated.
EG	11. If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
EG	12. ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4**SIMULATOR SCENARIO REVIEW CHECKLIST**

Note: The quantitative attribute target ranges that are specified on the form are not absolute limitations; some scenarios may be an excellent evaluation tool, but may not fit within the ranges. A scenario that does not fit into these ranges shall be evaluated to ensure that the level of difficulty is appropriate. (ES-301 Section D.5.d)

Initial	Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)	
GG	8	Total malfunctions inserted: 5-8
GG	2	Malfunctions that occur after EOP entry: 1-2
GG	3	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	1	EOPs entered/requiring substantive actions: 1-2
GG	1	EOP contingencies requiring substantive actions: 0-2
GG	2	Crew Critical Tasks: 2-3

COMMENTS:

ATTACHMENT 5
CRITICAL TASKS**11-01 NRC ESG-3**

CT #1 (TRIP-2 Basis) Establish the minimum required AFW flow (22E4 lbm/hr) prior to a transition to FRHS-1 being required.

Bases: Failure to establish the minimum required AFW flow for the secondary heat sink degrades the heat removal capability of the plant. Because the secondary heat sink is required but not satisfactorily provided, the RCS will heat up. Eventually, the core cooling CSF will be challenged. Ultimately, the fuel matrix/clad (a fission product barrier) will be challenged. (See PWROG Critical Task Book E-0-F)

CT#2 (FR-H.1-B) Establish RCS Bleed and Feed before PZR PORVs open automatically.

BASIS: Failure to establish RCS bleed and feed before automatic opening of the PORVs reduces the probability of success to establish a heat sink for the core. This constitutes a "significant reduction of the safety margin beyond that irreparably introduced by the scenario"

ATTACHMENT 6

ESG-PSA RELATIONSHIP EVALUATION

EVENTS LEADING TO CORE DAMAGE

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
N	TRANSIENTS with PCS Unavailable	N	Loss of Service Water
N	Steam Generator Tube Rupture	N	Loss of CCW
N	Loss of Offsite Power	N	Loss of Control Air
N	Loss of Switchgear and Pen Area Ventilation	N	Station Black Out
N	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
N	Containment Sump Strainers	N	Gas Turbine
N	SSWS Valves to Turbine Generator Area	N	Any Diesel Generator
N	RHR Suction Line valves from Hot Leg	Y	Auxiliary Feed Pump
N	CVCS Letdown line Control and Isolation Valves	N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
N	Cooldown the RCS and depressurize the system
N	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
Y	Initiate feed and bleed